

Developing training courses on Aland Combinatorial Optimisation for the ENTIRE EDIH



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Key Points

- Three courses on AI and optimisation
- Intended for SME and public PSO
- One or two day (scheduling), in person courses
- Focused on what can be achieved
- Mainly using case studies from Insight/UCC
- Developing open-source demonstrators for different applications
- Feedback from attendees very positive
- But, hard to reach target audience

ENTIRE EDIH



- European Digital Innovation Hubs (EDIH) program
- ENTIRE focused on Munster
- Funded by European Commission through Enterprise Ireland
- Provide upskilling for SME and PSO through "de minimis" funding



Scheduling and Planning

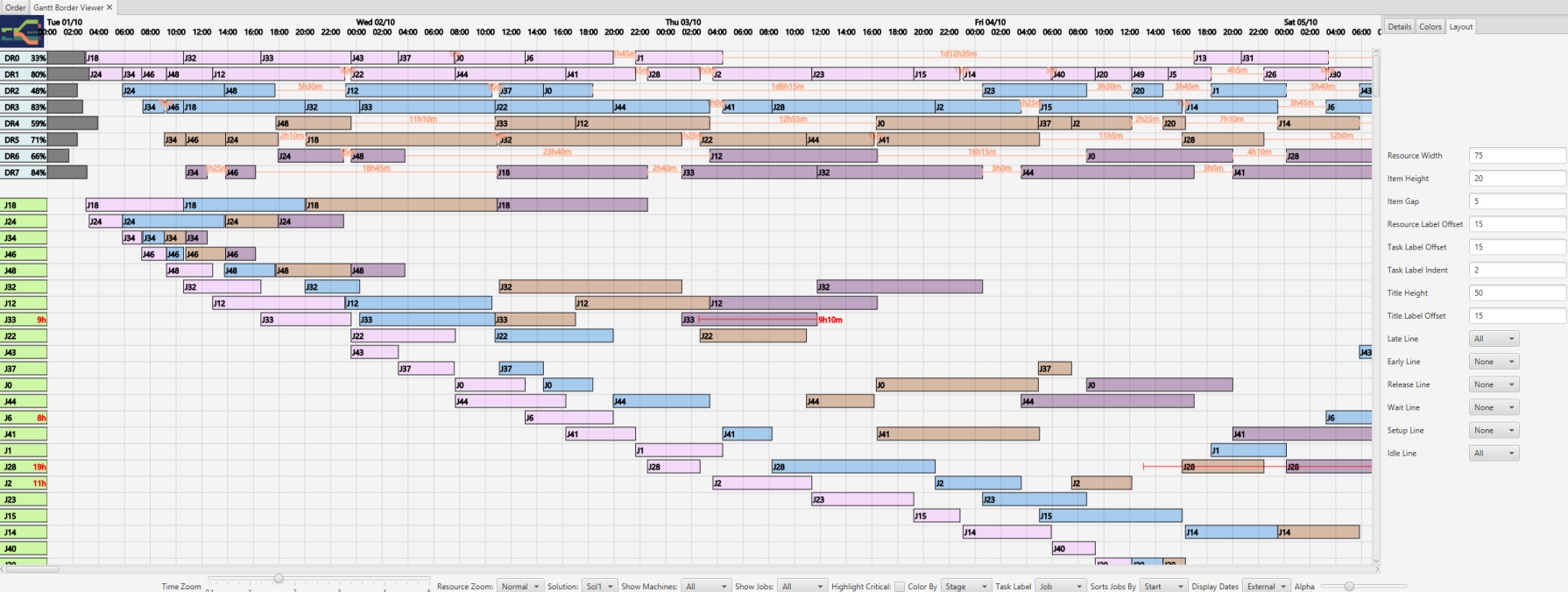
Topics

- Background/Motivation
- Scheduling Concepts
- Machine Constraints
- Costs and Objectives
- Advanced Concepts
- Case Studies
- Literature Review

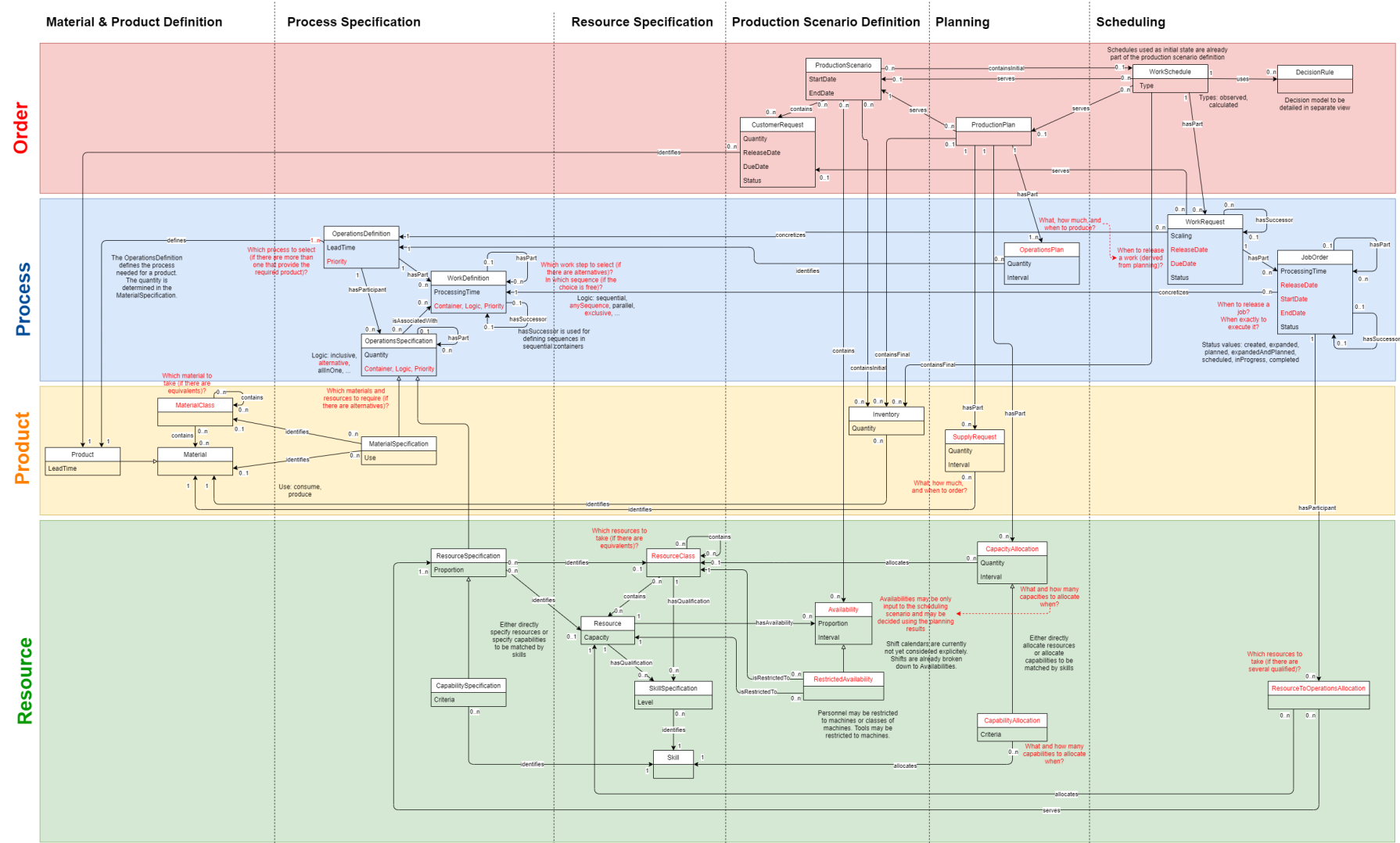
Scheduling Tool

- Open-source, generic scheduling tool
- Problem data/results as JSON file
- Connects to commercial and open-source solvers
- Programmed in Java and JavaFx

Gantt Chart Display



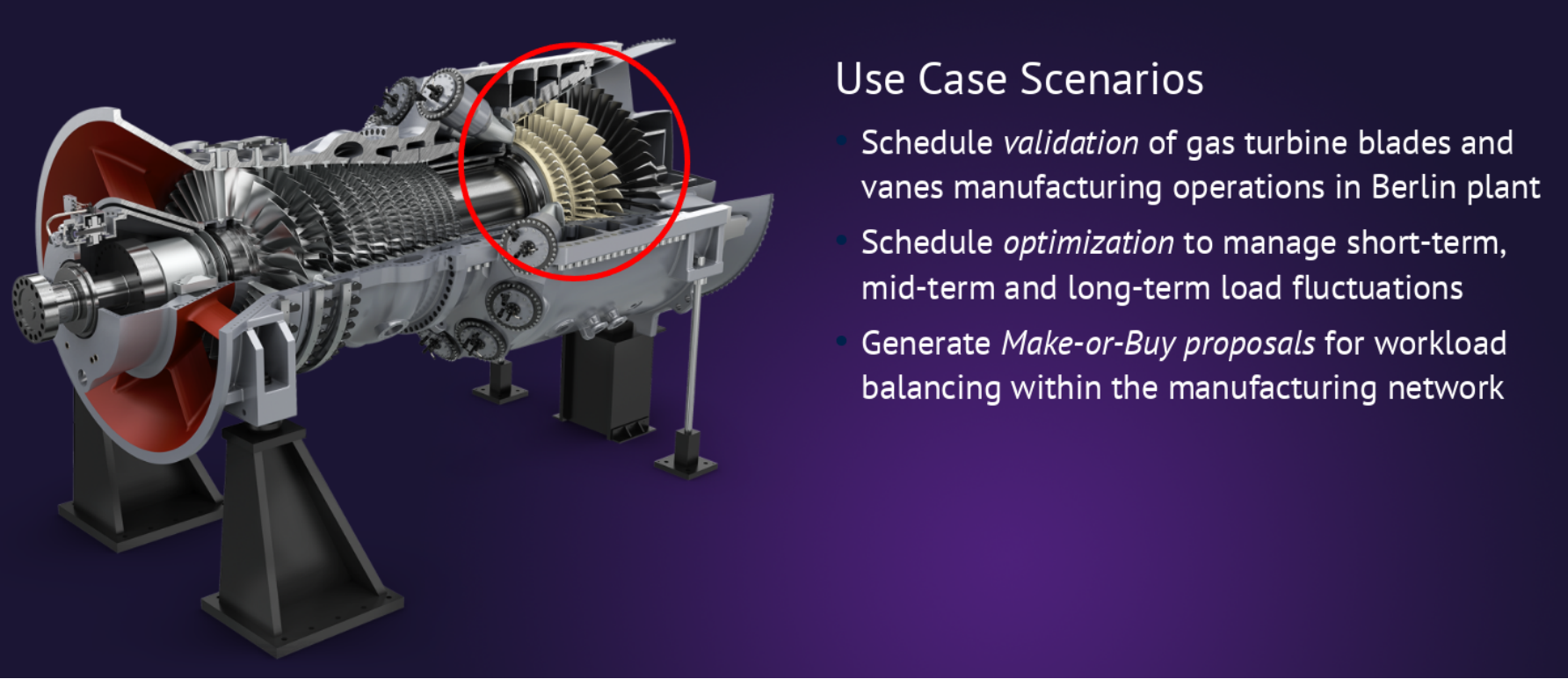
Underlying Domain Model



Case Studies

- Production Planning
 - Using Scheduling as a module
- Oven Scheduling
 - Unconventional resource constraints
- Blades and Vanes Production Planning
 - Multi-year scheduling for capacity planning

Blades and Vanes Problem



Use Case Scenarios

Schedule *validation* of gas turbine blades and vanes manufacturing operations in Berlin plant
Schedule *optimization* to manage short-term, mid-term and long-term load fluctuations
Generate *Make-or-Buy* proposals for workload balancing within the manufacturing network

Rostering and Timetabling

Topics

- Introduction
- AI and Constraint Programming
- Rostering
- Timetabling
- Case Studies
- Literature Survey

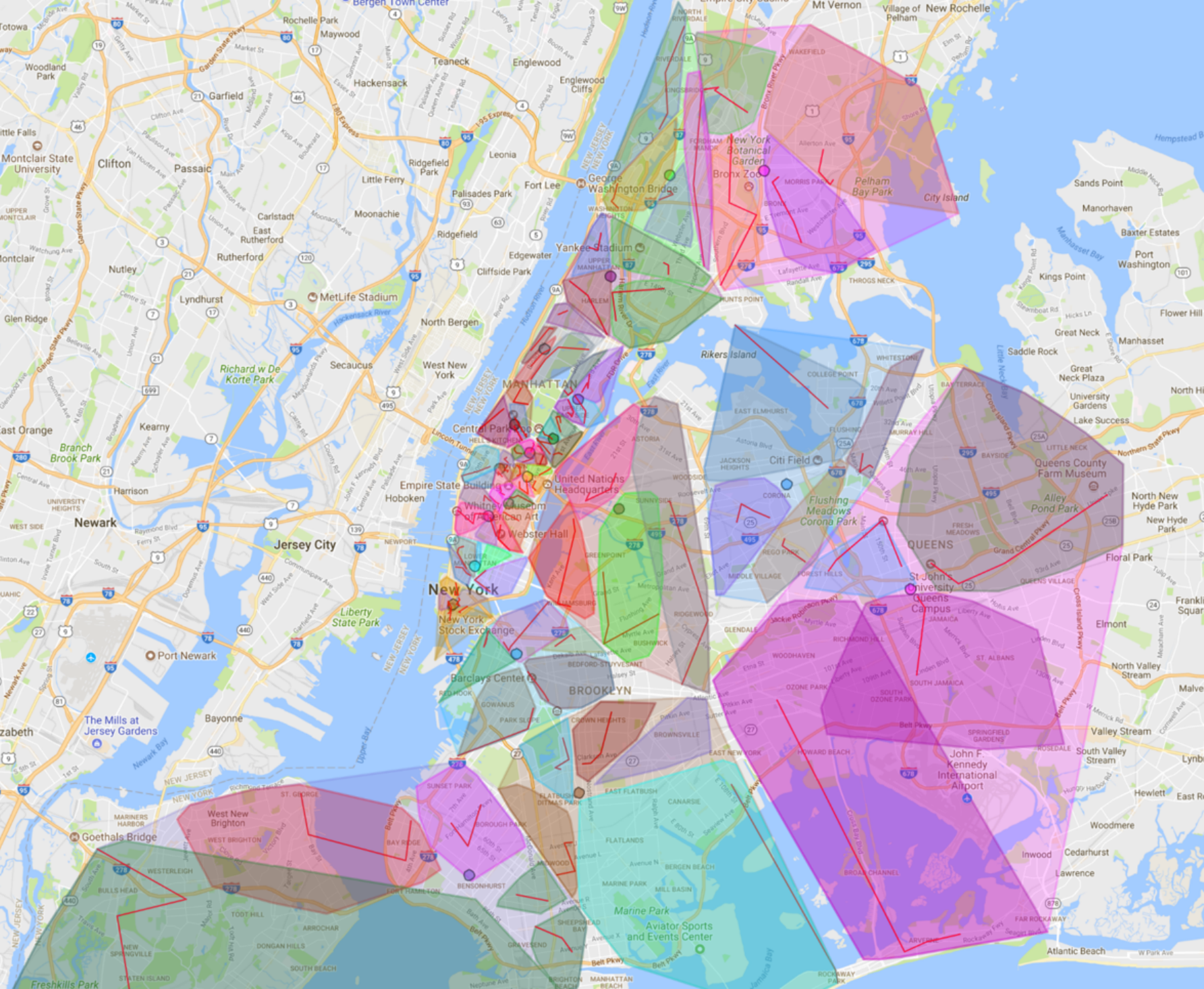
UCC's Dental School Timetabling



Case Studies

- Outpatient Waitlist Management
- Elevator Maintenance Planning and Scheduling
- Car Transportation
- Yearly Rostering for Irish Naval Service
- Catering Crew for TGV
- Rostering for TV/Radio Stations

Clustering and Route Planning

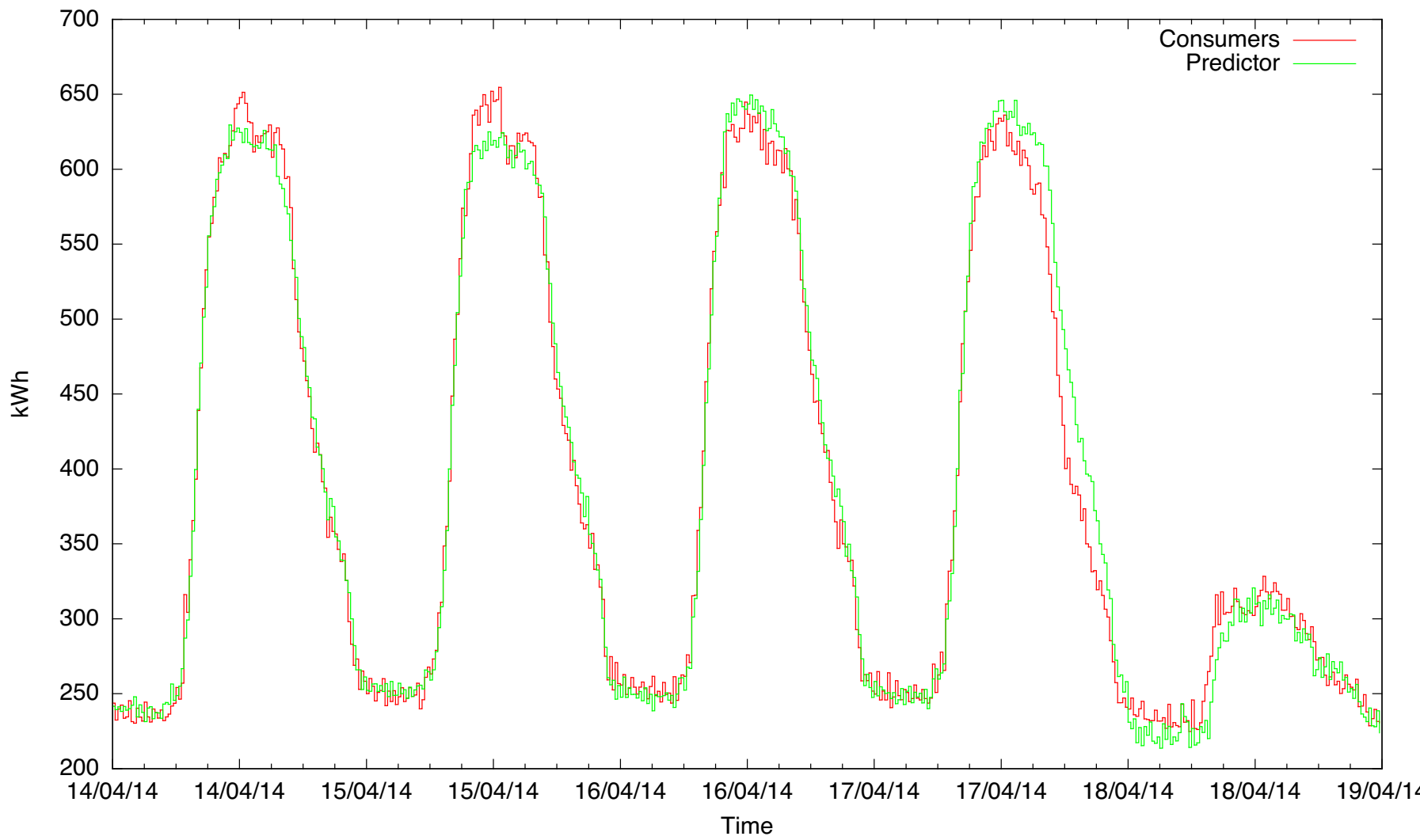


Energy Cost Optimisation

Topics

- Irish/EU Energy Market
- Data Analysis
- Load and Price Prediction
- Exploiting Renewables
- Literature Survey

Predicting UCC's Electricity Consumption



Case Studies

- Anomaly detection in induction motors
- Load forecasting in data centres
- SEM/I-SEM Electricity price prediction
- Scheduling UCC's CHP plant
- Feed-mill scheduling for minimising energy cost
- Tara mines dewatering

Mine Dewatering Problem



Course Delivery

- Scheduling (3x)
- Energy Cost Optimisation (2x)
- Rostering and Timetabling (once)

Planning repeat runs in 2026

- One-day version of scheduling course
- Split roosting and timetabling into separate courses

To learn more

ENTIRE Website

